

CTS SRE Day 13

Toil Reduction and Self-Healing Systems 1 & 2

EMPOWERING High Performance
Technology Teams

Agenda for the Day

- Introduction to Toil Reduction & Self-Healing Systems
- Identifying Toil in Operational Workflows
- Measuring the Impact of Toil
- Strategies for Toil Reduction Using Automation
- Automating Repetitive Tasks with Python/Cloudformation
- Building Self-Healing Systems for Resilience
- Hands-on Implementation: Self-Healing Kubernetes Containers
- Automated Monitoring with Prometheus Alerting Rules
- Summary & Key Takeaways

Introduction to Toil Reduction & Self-Healing Systems

- **Toil:** Manual, repetitive, and operationally expensive work that does not scale.
- **Goal:** Reduce human intervention, improve system reliability, and enhance efficiency.
- **Self-Healing Systems:** Automated recovery from failures without human intervention.

Identifying Toil in Operational Workflows

- Toil Characteristics:
 - Repetitive and predictable
 - Automatable but still done manually
 - Interrupts high-value work
 - Scales linearly with service growth
- Common Examples:
 - Manual log analysis
 - Restarting failed services
 - Running health checks

Measuring the Impact of Toil

- **Key Metrics to Track:**
 - % of time spent on repetitive tasks
 - Mean Time to Recovery (MTTR) trends
 - Number of alerts requiring manual intervention
- **Business Impact:**
 - Slower incident resolution
 - Increased human error risk
 - Reduced innovation time

Strategies for Toil Reduction Using Automation

- Automate Response Mechanisms (e.g., Auto-scaling, Event-driven execution)
- Standardize Runbooks & Scripting (e.g., Python, Ansible)
- Implement ChatOps for Operational Actions
- Leverage AI/ML-based Anomaly Detection

Identify Repetitive Tasks & Automate

- Common Manual Tasks:
 - Log analysis and anomaly detection
 - Configuration drift correction
 - System health checks
- **Automating with Python & Ansible:**
 - Use Python scripts for log analysis and auto-responses.
 - Implement Ansible playbooks for infrastructure automation.

Building Self-Healing Systems

- **Core Concept:** Systems that detect failures and recover autonomously.
- **Key Components:**
 - Automated recovery mechanisms
 - Predefined fallback strategies
 - Intelligent monitoring & alerting
- **Examples:**
 - Restarting failed containers in Kubernetes
 - Auto-scaling unhealthy instances in AWS

Implementing a Self-Healing Kubernetes System

- Problem: Containers crash, causing service downtime.
- Solution: Kubernetes Liveness & Readiness Probes
- Demo:

```
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
spec:
  containers:
    - name: example-container
      image: nginx
      livenessProbe:
        httpGet:
          path: /health
          port: 80
        initialDelaySeconds: 3
        periodSeconds: 5
```

Automate Monitoring Responses Using Prometheus

- Use Case: Automatically scale instances when CPU utilization exceeds a threshold.
- Steps:

1. Define Prometheus Alerting Rules:

```
groups:  
  - name: CPUAlerts  
    rules:  
      - alert: HighCPUUsage  
        expr: cpu_usage > 80  
        for: 5m  
        labels:  
          severity: critical
```

2. Connect Alerts to Auto-Scaling Actions.
3. Trigger Self-Healing via Kubernetes or AWS Lambda.

Hands on Exercises

Hands on exercises

- <https://www.eksworkshop.com/docs/introduction>

Lab Doc:

[EKS-self-healing script-Toil Reduction and Self-Healing Systems .docx](#)
[Python Script to Check System Logs - Toil Reduction and Self-Healing Systems .docx](#)

- Github Link:

<https://github.com/AshishSharmaPrivate/Lab/blob/main/EKS-self-healing%20script-Toil%20Reduction%20and%20Self-Healing%20Systems>

- <https://github.com/AshishSharmaPrivate/Lab/blob/main/Python%20Script%20to%20Check%20System%20Logs%20-%20Toil%20Reduction%20and%20Self-Healing%20Systems>

Summary & Key Takeaways

- ✓ **Toil Reduction:** Identify and eliminate repetitive operational tasks.
- ✓ **Automation Strategies:** Use scripting, AI-based detection, and automation tools.
- ✓ **Self-Healing Systems:** Implement failure recovery using Kubernetes & Prometheus.
- ✓ **Long-Term Benefits:** Faster issue resolution, reduced manual intervention, improved reliability.

Thank You!
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